

One-day-ahead 1% VaR forecasting on QQQ daily log returns

Comparison of Multiple model classes

Ethan Yao

Project Objective

Build and compare a reproducible VaR forecasting framework that combines:

- Rolling out-of-sample forecasts
- Multiple model classes (parametric, semi-parametric, nonparametric, direct quantile)
- Statistical backtesting
- Plot diagnostics

The real goal is not just in-sample fit. The goal is model comparison under realistic tail-risk forecasting conditions.

Data and sample

Dataset:

- QQQ daily adjusted close
- VIX close

Construction:

- Compute daily log returns from QQQ close price: $r_t = \log\left(\frac{P_t}{P_{t-1}}\right)$
- VIX used as exogenous regressor for GARCHX
- Data pipeline starts at 2015-01-01

Current local files indicate:

- Full modeled sample runs roughly from 2015-01-02 to 2026-03-27
- Initial training fraction: 70%
- Out-of-sample period used in saved backtests: 848 trading days
- Pipeline forecasts start around 2022-11-01

Setup and notation

Let r_t denote the daily log return at time t . In several implementations, the code works with a scaled series $y_t = 100 r_t$ for numerical stability when fitting volatility models.

For one-step-ahead Value-at-Risk at level α , the target object is the left-tail conditional quantile $\text{VaR}_{t+1}(\alpha) = q_{t+1|t}(\alpha)$, so that ideally

$$\Pr(r_{t+1} < \text{VaR}_{t+1}(\alpha) \mid \mathcal{F}_t) = \alpha.$$

Gaussian or Student-(t) tail

- If `dist="normal"`, then

$$q_{\alpha} = \Phi^{-1}(\alpha)$$

- If `dist="t"`, the code uses a Student-(t) quantile with estimated degrees of freedom (ν), then rescales it to unit variance by multiplying by

$$\sqrt{\frac{\nu - 2}{\nu}}$$

So the final one-step forecast is

$$\text{VaR}_{t+1}(\alpha) = \mu_{t+1} + \sigma_{t+1} q_{\alpha}$$

When the model is conditionally location-scale, the generic structure is

$$r_{t+1} = \mu_{t+1} + \sigma_{t+1} z_{t+1},$$

which implies

$$\text{VaR}_{t+1}(\alpha) = \mu_{t+1} + \sigma_{t+1} q_{\alpha},$$

where q_{α} is either a parametric quantile (Gaussian or Student- t) or an empirical residual quantile, μ is the mean comes from the fitted `arch_model(...).fit()` result, σ_{t+1}^2 comes from the one-step-ahead conditional variance forecast through `arch_model.fit.forecast(horizon=1, reindex=False).variance`.

Model Suite

0. Benchmark Model

- RiskMetrics with $\lambda = 0.94$ and initialized by variance estimate from previous 250 observations

1. Original pipeline models

- GARCH(1,1) - Gaussian/Student-t
- Nonparametric Quantile-GARCH SD - Gaussian/Student-t
- Nonparametric-GARCH Bootstra - Gaussian/Student-t

2. Extension GARCH models

- GARCHX(VIX)
- GJR-GARCH - Gaussian/Student-t

3. Direct Simulation Methods

- CAViaR-SAV
- Historical Simulation

* Financial returns usually have fatter tails than a Gaussian distribution. Gaussian model says extreme moves are very rare, but in real financial data, large losses and gains happen more often than Gaussian theory predicts. The Student-t distribution handles this better because it has heavier tails.

Model formulas / interpretation

RiskMetrics Model

$$r_t = \sigma_t W_t, W_t \sim N(0, 1)$$

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) t - 1^2$$

$$VaR_{t,1}(\alpha) = \left(1 - e^{\sigma_{t+1} \Phi^{-1}(\alpha)}\right) p_t$$

GARCH(1,1)

Variance recursion:

$$x_t = \sigma_t W_t, W_t \sim N(0, 1) / Student - t$$

$$\sigma_t^2 = \omega + a x_{t-1}^2 + b \sigma_{t-1}^2,$$

VaR computed from one-step conditional volatility and a parametric quantile.

GARCHX(VIX)

Variance recursion:

$$\sigma_t^2 = \omega + a \times \epsilon_{t-1}^2 + b \times \sigma_{t-1}^2 + g \times X_{t-1}$$

where X_t is a transformed VIX regressor.

Current implementation now uses:

- returns scaled by 100 for optimization stability
- VIX standardized, shifted positive, then squared inside the variance recursion
- Gaussian quasi-likelihood for the GARCHX fit itself
- L-BFGS-B optimization with positivity bounds
- fallback to plain GARCH when optimization fails

This revision is meant to replicate the GARCHX mechanics first. Performance can be evaluated later.

GJR-GARCH

Variance recursion:

$$\sigma_t^2 = \omega + a \times \epsilon_{t-1}^2 + \gamma \times \epsilon_{t-1}^2 \times I(\epsilon_{t-1} < 0) + b \times \sigma_{t-1}^2$$

This is the asymmetric / leverage-effect extension.

In the code, asymmetry is handled by `arch_model(..., p = 1, o = 1, q = 1, ...)`.

Direct Quantile Forecasts

CAViaR-SAV

Paper idea: - CAViaR models the conditional quantile directly rather than modeling volatility first. - The Symmetric Absolute Value (SAV) specification in Engle-Manganelli uses lagged quantile plus lagged absolute return. - In the paper, the quantile process is estimated by regression quantiles and model adequacy is assessed with the Dynamic Quantile (DQ) test.

Quantile recursion in the current code is written on the positive VaR magnitude:

$$v_t = \beta_0 + \beta_1 \times v_{t-1} + \beta_2 \times |r_{t-1}|$$

and then mapped back to the left-tail return quantile with

$$q_t = -v_t$$

with returns scaled by 100 inside optimization.

The remaining practical flaw is weak optimization: The current code still uses only a small set of starting values, Nelder-Mead only, and a fairly light iteration budget. Relative to the paper, this is a weak search strategy and can easily leave the fit stuck in a poor local solution.

What aligns with the paper:

- direct conditional quantile modeling rather than volatility-first modeling
- SAV structure driven by lagged quantile and lagged absolute return
- quantile loss / regression-quantile objective
- Nelder-Mead optimization with multiple starts

What does not yet fully match the paper:

- no Dynamic Quantile (DQ) test in the current project runner
- only the SAV variant is implemented; Adaptive, Asymmetric Slope, and Indirect GARCH CAViaR are not
- the optimizer setup is simpler than the paper's broader search routine and does not include the paper's quasi-Newton refinement step
- initialization is simplified relative to the paper's empirical quantile

Historical Simulation

Rolling empirical quantile of the most recent window of returns. Current base window: 250 days.

Nonparametric Quantile-GARCH SD/Bootstrap

GARCH-FHS

Filtered Historical Simulation:

- fit GARCH
- compute standardized residuals
- take empirical alpha-quantile of residuals
- scale by $\hat{\sigma}_{t+1}$

GJR-FHS Same empirical residual-quantile idea as FHS, but applied after fitting GJR-GARCH.

GARCH-Bootstrap One-step residual bootstrap: - fit GARCH - resample standardized residuals - simulate one-step return - take empirical alpha-quantile of simulated returns For one-step forecasting this is very close to FHS in practice.

Backtesting framework

At each out-of-sample date: 1. Fit model on data up to time $t-1$ 2. Forecast one-step VaR for time t 3. Compare realized return r_t to forecast VaR_t 4. Record violation indicator:

$$I_t = 1(r_t < VaR_t)$$

Backtests currently reported in the pipeline:

- Kupiec unconditional coverage test
- Christoffersen independence test
- Conditional coverage test

Important note:

- These are standard backtests and are useful.
- However, for CAViaR specifically, the reference paper emphasizes the Dynamic Quantile (DQ) test as the model adequacy diagnostic.
- So the current project can say the CAViaR implementation is improved and

Lecture 18 alignment

Current status relative to Stat 929 Lecture 18:

Strong alignment now: - Historical Simulation - GARCH-Gaussian - Empirical residual quantile / FHS style GARCH - Residual bootstrap GARCH - Rolling backtesting architecture

Broad alignment but more project-specific: - GARCH Student-t - GJR-GARCH - GJR-FHS - GARCHX(VIX)

Still outside lecture core: - CAViaR-SAV

Bottom line: The extension models brought the project much closer to the lecture's VaR forecasting framework. The biggest lecture-consistent success is the empirical-tail GARCH variants.

Key implementation fixes already made

A. GARCHX fixes - returns scaled by 100 for optimization stability - more sensible starting values and bounds - explicit raw-return fallback to standard GARCH - transformed VIX regressor for numerical stability - stationarity constraint enforced through SLSQP

CAViaR fixes

- returns scaled by 100 for optimization stability
- SAV recursion is now documented using positive VaR magnitude and then mapped back to a negative return quantile
- multiple starts are still used
- however, optimization is still weak relative to the paper: few starts, Nelder-Mead only, and limited iterations
- a proper upgrade would add a broader search and a post-fit adequacy check such as a Dynamic Quantile style diagnostic

Extension model fixes

- added lecture-aligned GARCH-Gaussian
- added FHS GARCH
- added FHS GJR
- added residual bootstrap GARCH
- separate extension runner and plotting script

Current saved backtest results

Alpha = 1% Out-of-sample observations per model = 846 Expected violations = 8.46

A. Original pipeline results

1. Historical-Simulation

- Violations: 12
- Violation rate: 1.42%
- Kupiec p-value: 0.250
- Christoffersen p-value: 0.157
- Conditional coverage p-value: 0.190 Verdict: strong benchmark; stable and defensible

2. GARCH-Gaussian

- Violations: 22
- Violation rate: 2.60%

Plot-based interpretation

Saved plots are consistent with the backtests: - Historical Simulation gives a stepwise, slow-moving VaR line but remains competitive - GARCH Student-t reacts to stress periods reasonably well - GJR-GARCH reacts to stress, but not as effectively as GJR-FHS - GARCHX looks visually similar to plain GARCH but backtests worse - GARCH-Gaussian sits too close to zero during stressed periods and gets breached too often - GARCH-FHS and GJR-FHS become more negative during volatility spikes and produce much better calibration - GARCH-Bootstrap is visually close to GARCH-FHS, which is expected for one-step forecasting - CAViaR-SAV in the saved results still sits far too close to zero and gets breached constantly

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